EMS611U: Conceptual Design of Aircraft

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Week 8

14 November 2025

Recap – Sizing from a conceptual sketch

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

Requirement parameters

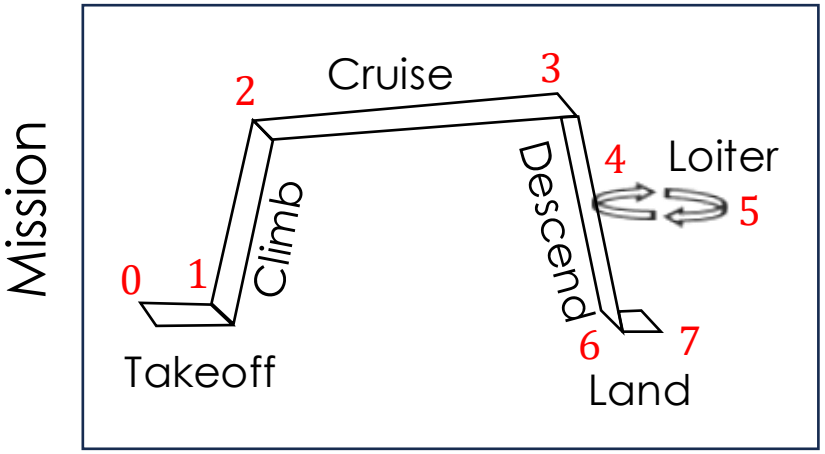
$$\frac{W_e}{W_0} = A\, W_0^C K_{vs}$$

$W_e/W_0 = A W_0^C K_{vs}$	A	C
Sailplane—unpowered	0.86	−0.05
Sailplane—powered	0.91	−0.05
Homebuilt—metal/wood	1.19	−0.09
Homebuilt—composite	0.99	−0.09
General aviation—single engine	2.36	−0.18
General aviation—twin engine	1.51	−0.10
Agricultural aircraft	0.74	−0.03
Twin turboprop	0.96	−0.05
Flying boat	1.09	−0.05
Jet trainer	1.59	−0.10
Jet fighter	2.34	−0.13
Military cargo/bomber	0.93	−0.07
Jet transport	1.02	−0.06

K_{vs} = variable sweep constant = 1.04 if variable sweep
= 1.00 if fixed sweep

Recap – Sizing from a conceptual sketch

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$



	(W_i/W_{i-1})
Warmup and takeoff	0.970
Climb	0.985
Landing	0.995

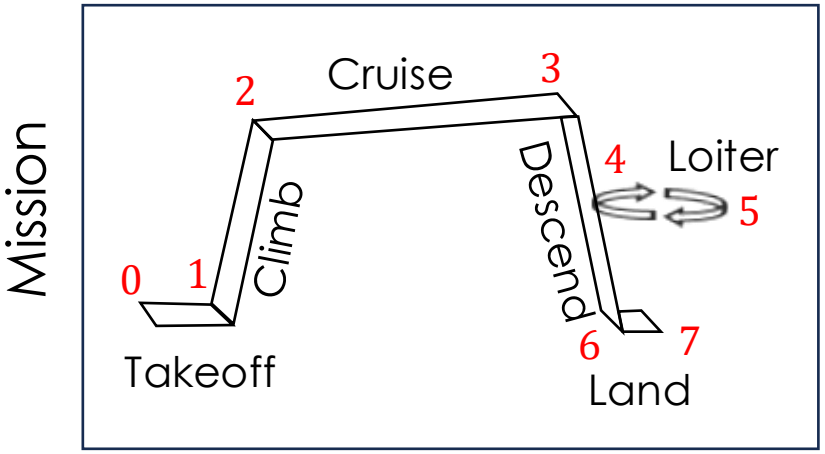
Mission fuel fraction

$$1 - \frac{W_{final}}{W_{initial}} = \frac{W_x}{W_0}$$

$$\frac{W_{final}}{W_{initial}} = \frac{W_7}{W_0} = \frac{W_1}{W_0}\frac{W_2}{W_1}\frac{W_3}{W_2}\frac{W_4}{W_3}\frac{W_5}{W_4}\frac{W_6}{W_5}\frac{W_7}{W_6}$$

Recap – Sizing from a conceptual sketch

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$



Cruise

$$\frac{W_i}{W_{i-1}} = e^{\frac{-RC}{V(L/D)}}$$

Loiter

$$\frac{W_i}{W_{i-1}} = e^{\frac{-EC}{L/D}}$$

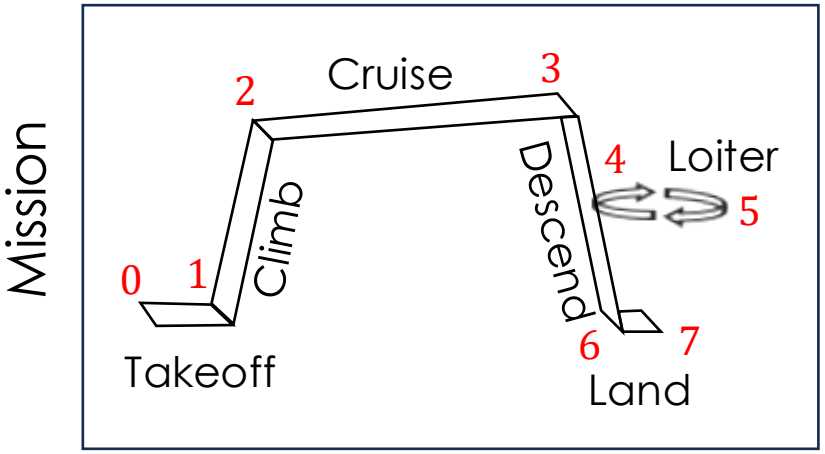
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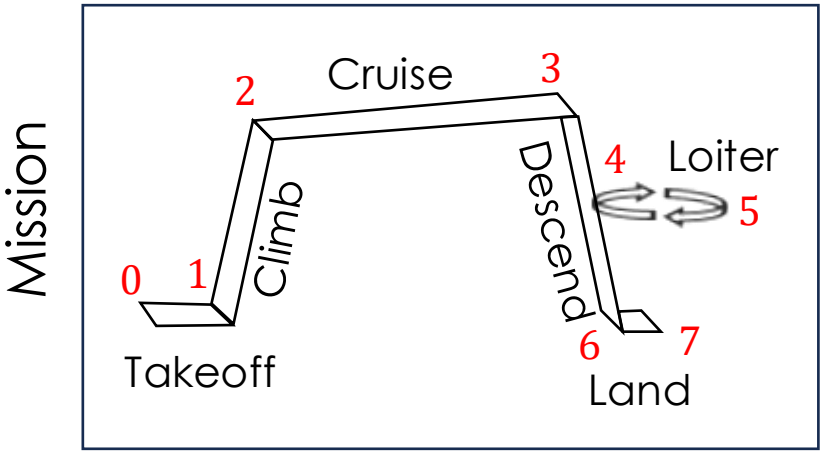
Typical jet SFC's	Cruise	Loiter
Pure turbojet	0.9	0.8
Low-bypass turbofan	0.8	0.7
High-bypass turbofan	0.5	0.4

Propeller: $C = C_{bhp} V/(550\eta_p)$

Typical C_{bhp} and η_p	Cruise	Loiter
Piston-prop (fixed pitch)	0.4/0.8	0.5/0.7
Piston-prop (variable pitch)	0.4/0.8	0.5/0.8
Turboprop	0.5/0.8	0.6/0.8

Recap – Sizing from a conceptual sketch

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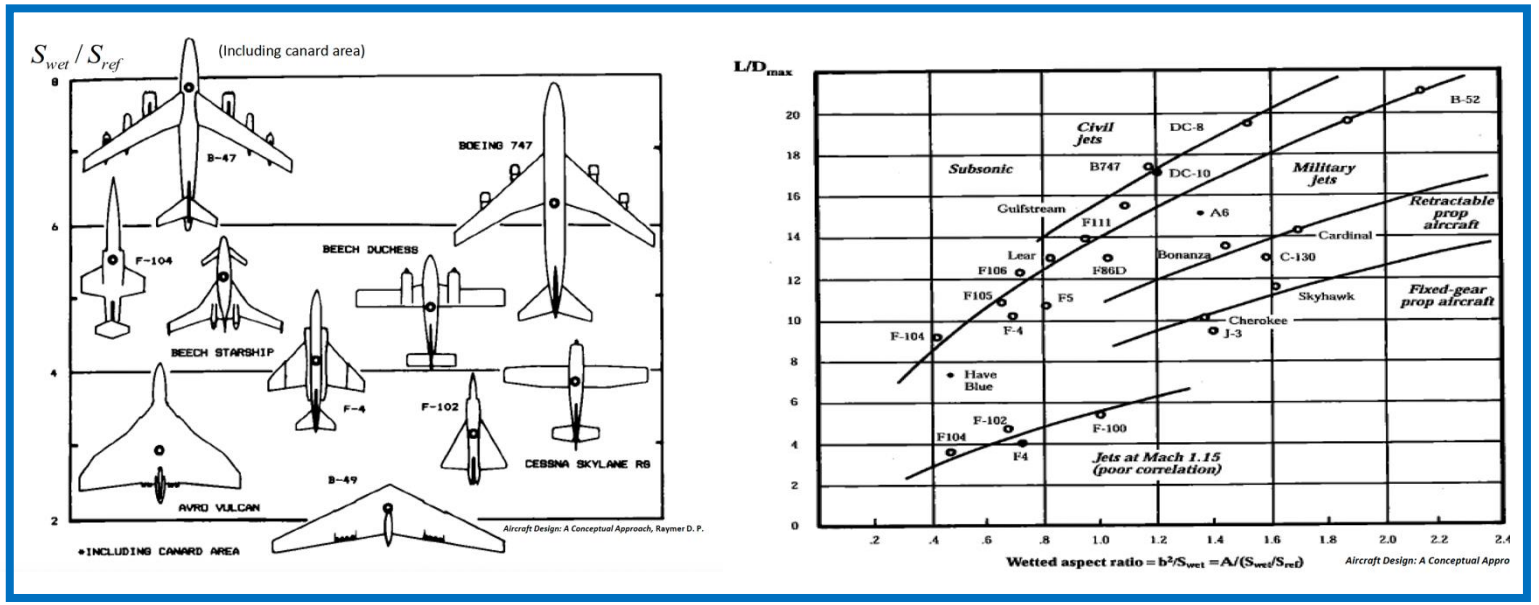


Cruise

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Mission fuel fraction

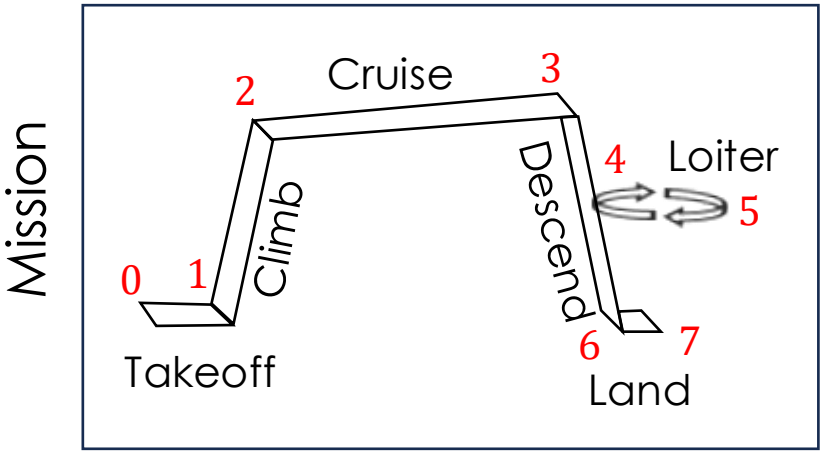
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	Cruise	Loiter
Jet	$0.866\,(L/D)_{max}$	$(L/D)_{max}$
Prop	$(L/D)_{max}$	$0.866\,(L/D)_{max}$

Recap – Sizing from a conceptual sketch

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$



Mission fuel fraction

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If %6 reserve fuel:

$$\frac{W_f}{W_0} = 1.06\left(1 - \frac{W_x}{W_0}\right)$$

Recap – Sizing from a conceptual sketch

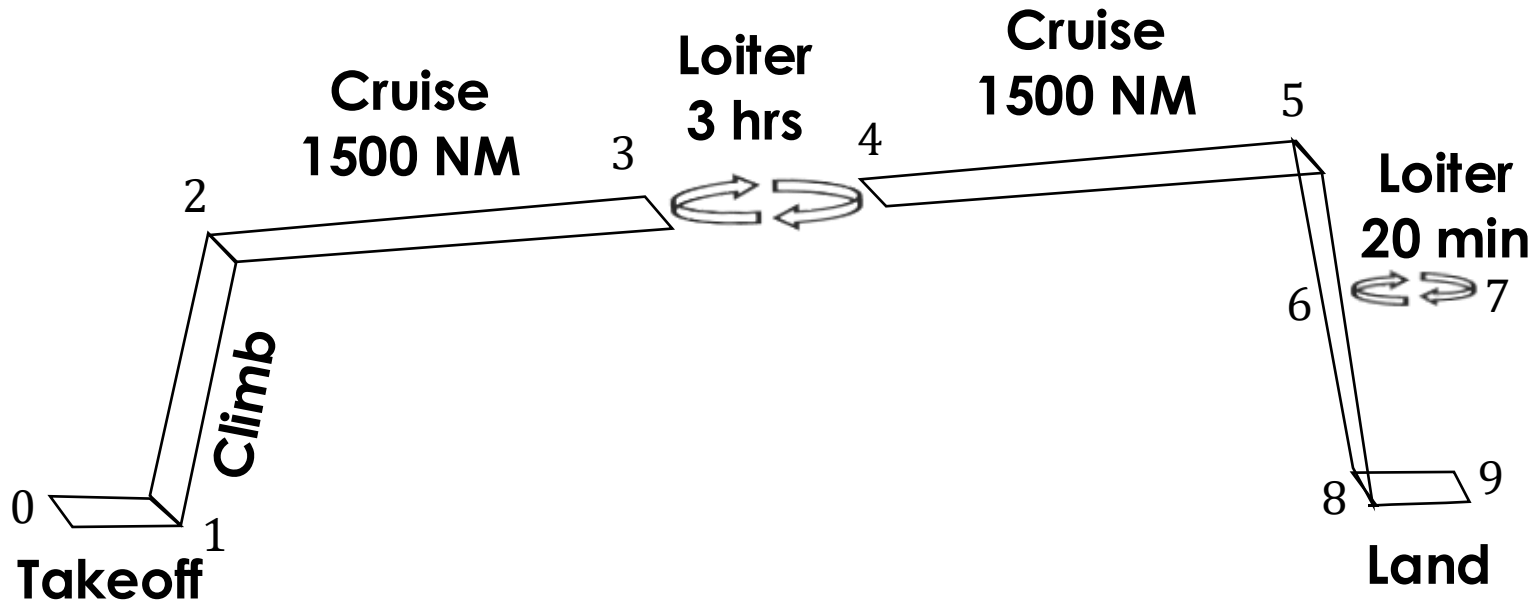
$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

- Guess W_0
- iterate

Design Example: ASW Aircraft

Requirements:

- Mission requirement for ASW (antisubmarine warfare) aircraft:



Requirements:

- This aircraft is expected to use sophisticated electronic equipment to detect and track submarines that weighs 10,000 lb.
- This aircraft is expected to carry 4 people crew, totaling 800 lb.
- The aircraft must cruise at 0.6 Mach Number.
- Equivalent combined aspect ratio of 7.

Design Example: ASW Aircraft

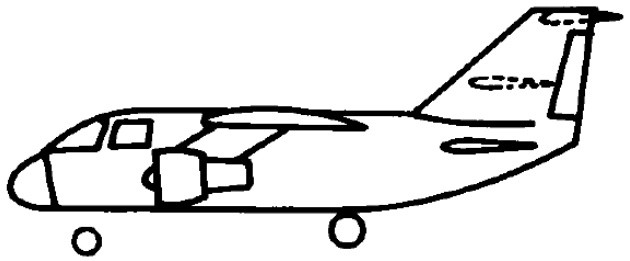
1. Come up with conceptual sketches
   - Looking at an aircraft that does a similar mission.

Lockheed S-3A



Design Example: ASW Aircraft

1. Come up with conceptual sketches
   - Looking at an aircraft that does a similar mission.

Conventional


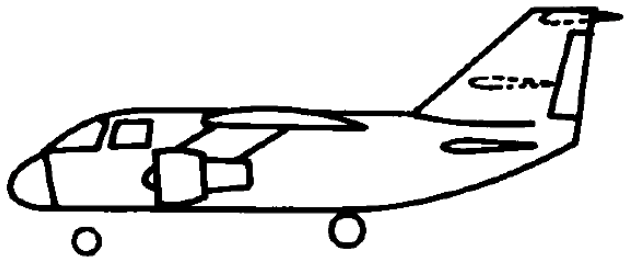
Conventional:

- The low horizontal tail position offers the lightest structure, but tail might be in the wake of the engine.
- So other positions of the tail are given with dashed line.

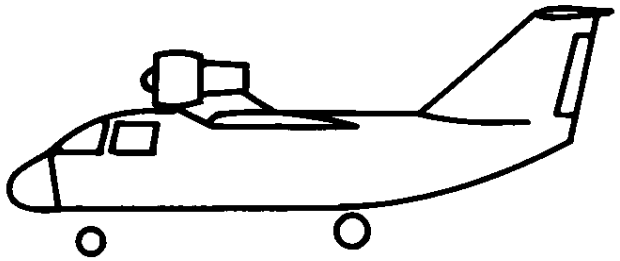
Design Example: ASW Aircraft

1. Come up with conceptual sketches
   - Looking at an aircraft that does a similar mission.

Conventional



Over-the-wing Nacelles



Over-the-wing Nacelles:

- Creates more lift due to engine position.
- Greater ground clearance for the engines, thereby reducing the tendency to suck-up debris.
- Disadvantage: Difficulty reaching the engines for maintenance.

Design Example: ASW Aircraft

1. Come up with conceptual sketches
   - Looking at an aircraft that does a similar mission.

Conventional

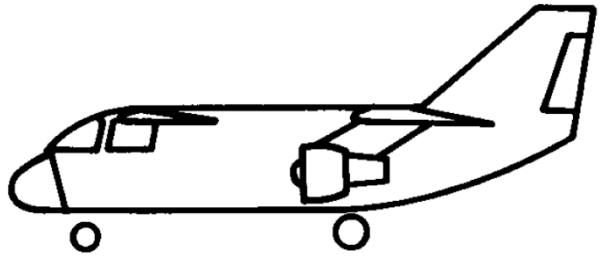
Canard – low wing

Canarded options:

- Canards offer benefits like reduced trim drag – wider allowable range of center of gravity.

Over-the-wing Nacelles

Canard – high wing

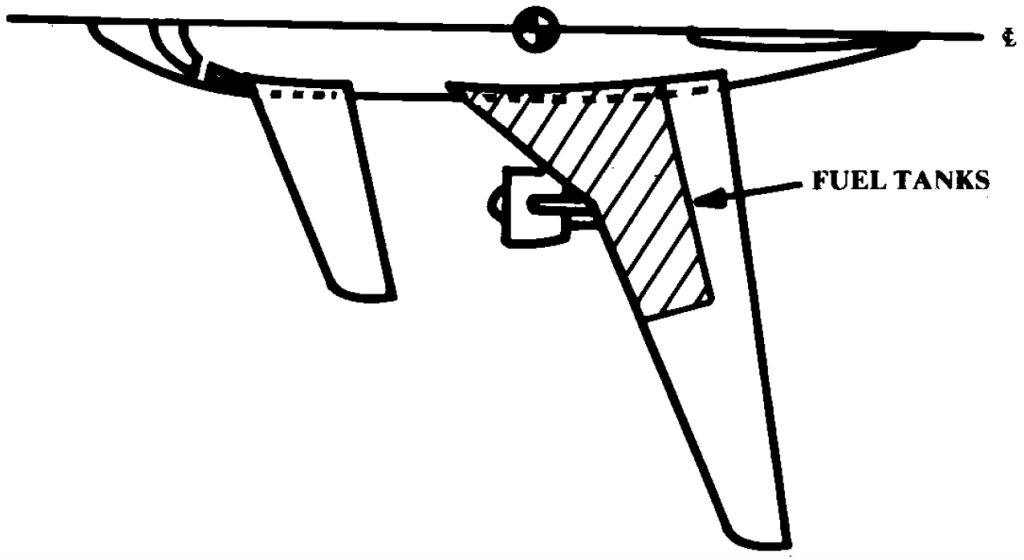


Canard – high wing:

- Easier for engine maintenence

Design Example: ASW Aircraft

Conceptual Sketch in more detail:



The fuel tanks should be placed so that the fuel is evenly distributed about the aircraft center of gravity (estimated location shown by the circle with two quarters shaded).

This is necessary so that the aircraft when loaded has nearly the same center of gravity as when its fuel is almost gone.

Problem:

The wing is located aft of the center of gravity whenever a canard is used, so, the fuel located in the wing is also aft of the center of gravity.

One solution to this problem would be to add fuel tanks in the fuselage, forward of the center of gravity. This would increase the risk of fire in the fuselage during an accident, and is forbidden in commercial aircraft.

Another solution is to add a wing strake full of fuel. The strakes do add to the aircraft wetted area, which reduces cruise aerodynamic efficiency.

Design Example: ASW Aircraft

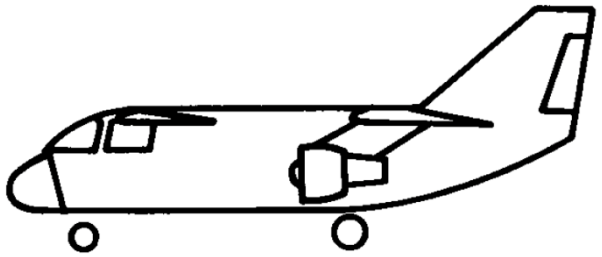
All aircraft design entails a series of trade offs.

- The canard offers lower trim drag, but in this case may require a higher wetted area. The only true way to determine whether a canard is a good idea for this or any aircraft is to design several aircraft, one with and one without a canard.

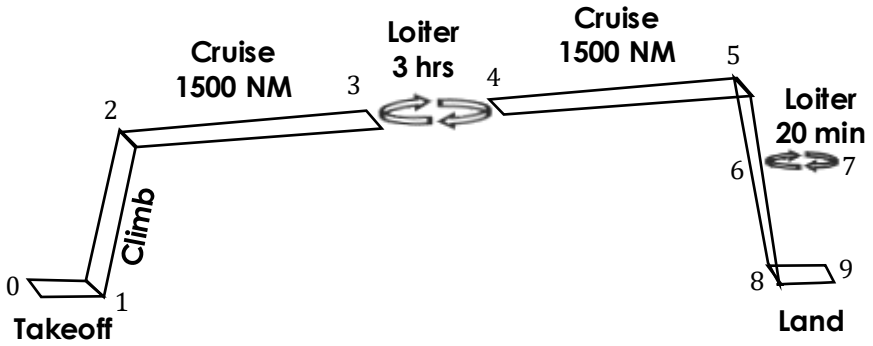
- This type of trade study comprises the majority of the design effort during the conceptual design process.

Design Example: ASW Aircraft

Conceptual Sketch:



Mission:



Requirements:

- This aircraft is expected to use sophisticated electronic equipment to detect and track submarines that weighs 10,000 lb.
- This aircraft is expected to carry 4 people crew, totaling 800 lb.
- The aircraft must cruise at 0.6 Mach Number.
- Equivalent combined aspect ratio of 7.

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

$$W_{crew} + W_{payload} = 10000 + 800 = 10800\ lb$$

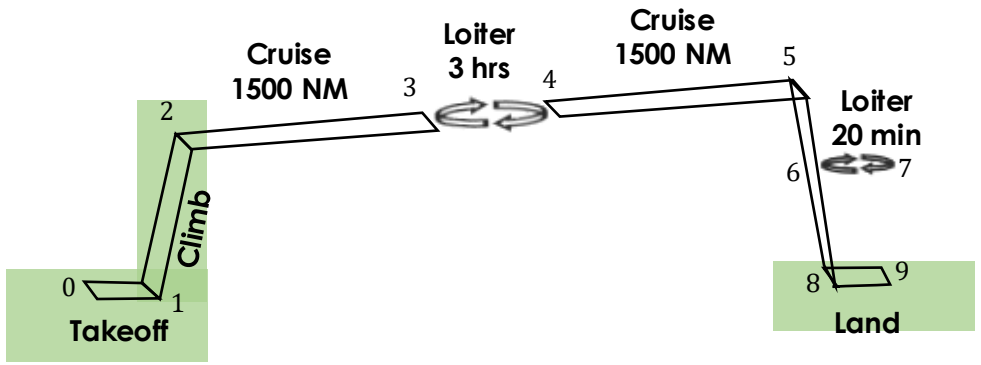
$$\frac{W_e}{W_0} = A\, W_0^C K_{vs} = 0.93\, W_0^{-0.07}$$

$W_e/W_0 = A W_0^C K_{vs}$	A	C
Military cargo/bomber	0.93	−0.07

K_{vs} = variable sweep constant = 1.04 if variable sweep
= 1.00 if fixed sweep

Design Example: ASW Aircraft

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$$1 - \frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}}$$ Mission fuel fraction

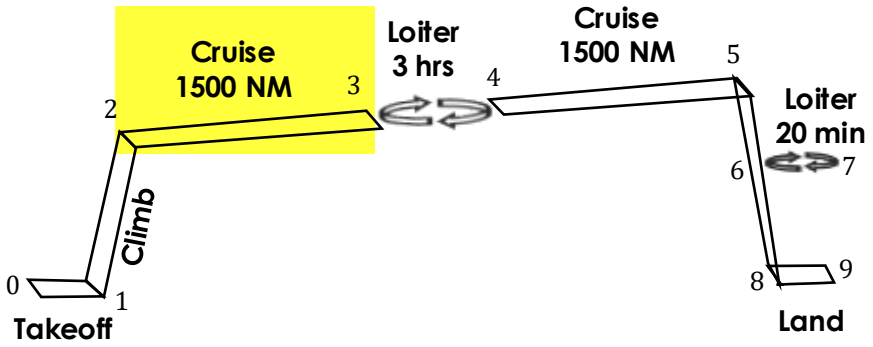
$$\frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}} = \frac{W_9}{W_0} = \frac{W_1}{W_0}\frac{W_2}{W_1}\frac{W_3}{W_2}\frac{W_4}{W_3}\frac{W_5}{W_4}\frac{W_6}{W_5}\frac{W_7}{W_6}\frac{W_8}{W_7}\frac{W_9}{W_8}$$

0.97 0.985 0.995

	(W_i/W_{i-1})
Warmup and takeoff	0.970
Climb	0.985
Landing	0.995

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$$\frac{W_3}{W_2} = e^{\frac{-RC}{V(L/D)}}$$

$R = 1500nm = 9114000\ ft$

$M = V/a$

$V = M * a = 0.6\left(994.8\frac{ft}{s}\right)$

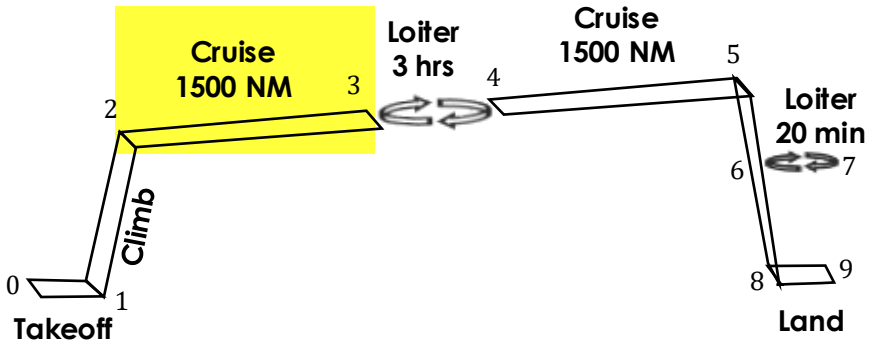
$= 569.9\ ft/s$

$C = 0.5\frac{1}{hr} = 0.0001389\frac{1}{s}$

Typical jet SFC's	Cruise	Loiter
Pure turbojet	0.9	0.8
Low-bypass turbofan	0.8	0.7
High-bypass turbofan	0.5	0.4

Design Example: ASW Aircraft

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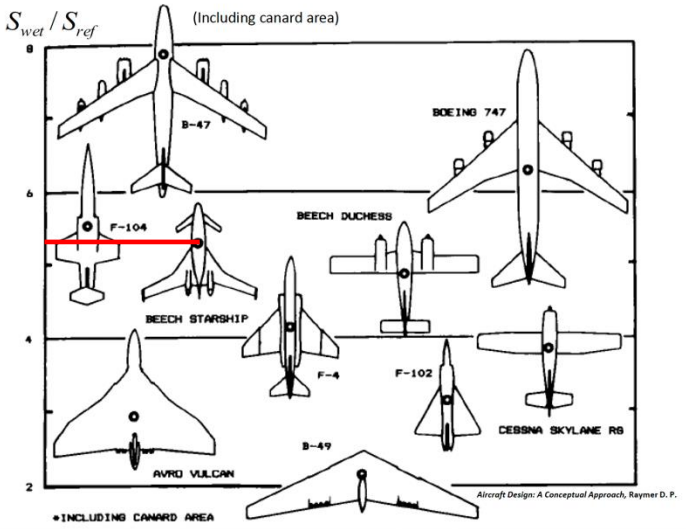
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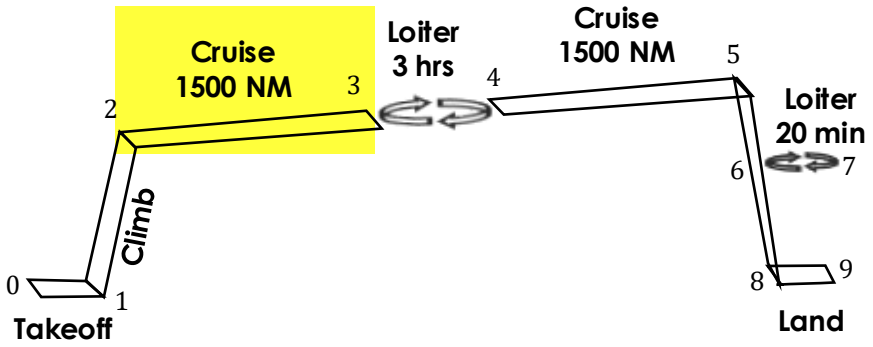
$$\frac{S_{wet}}{S_{ref}} \approx 5.5$$

$$AR = 7$$

$$\frac{AR}{\frac{S_{wet}}{S_{ref}}} = \frac{7}{5.5} = 1.27$$

Design Example: ASW Aircraft

Mission:



- The aircraft must cruise at 0.6 Mach Number.
- Equivalent combined aspect ratio of 8.

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

$$1 - \frac{W_{final}}{W_{initial}}$$ Mission fuel fraction

$$\frac{W_{final}}{W_{initial}} = \frac{W_9}{W_0} = (0.97)(0.985)\frac{W_3}{W_2}\frac{W_4}{W_3}\frac{W_5}{W_4}\frac{W_7}{W_6}(0.995)$$

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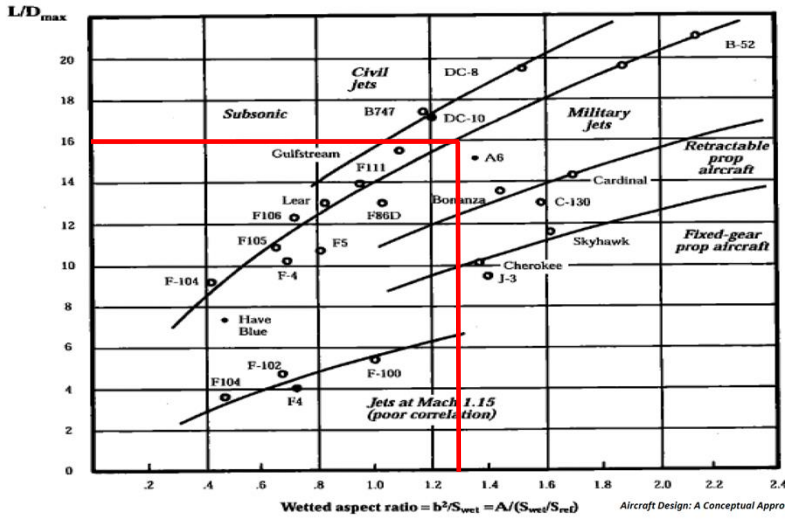
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$= 569.9\ ft/s$

$C = 0.5\frac{1}{hr} = 0.0001389\frac{1}{s}$



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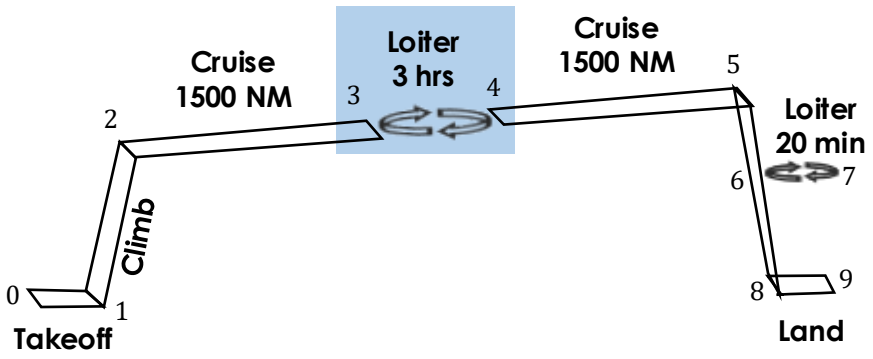
$\frac{AR}{\frac{S_{wet}}{S_{ref}}} = \frac{7}{5.5} = 1.27$

$\left(\frac{L}{D}\right)_{max} \approx 16$ $\left(\frac{L}{D}\right) = 0.866\left(\frac{L}{D}\right)_{max} = 13.9$

	Cruise
Jet	$0.866\,(L/D)_{max}$

Design Example: ASW Aircraft

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$$1 - \frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}}$$ Mission fuel fraction

$$\frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}} = \frac{W_9}{W_0} = (0.97)(0.985)\,(0.853)\frac{W_4}{W_3}\frac{W_5}{W_4}\frac{W_7}{W_6}(0.995)$$

$$\frac{W_4}{W_3} = e^{\frac{-EC}{(\boldsymbol{L/D})}}$$

$E = 3\ hr = 10800\ s$

$C = 0.4\frac{1}{hr} = 0.000111\frac{1}{s}$

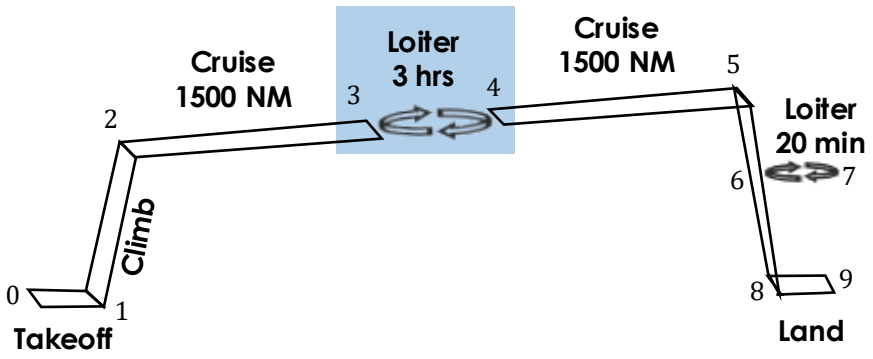
$\left(\frac{L}{D}\right) = \left(\frac{L}{D}\right)_{max} = 16$

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	Cruise	Loiter
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$$\frac{W_4}{W_3} = e^{\frac{-EC}{(\boldsymbol{L/D})}} = e^{\frac{-10800(0.00011)}{(16)}} = 0.9277$$

$E = 3\ hr = 10800\ s$

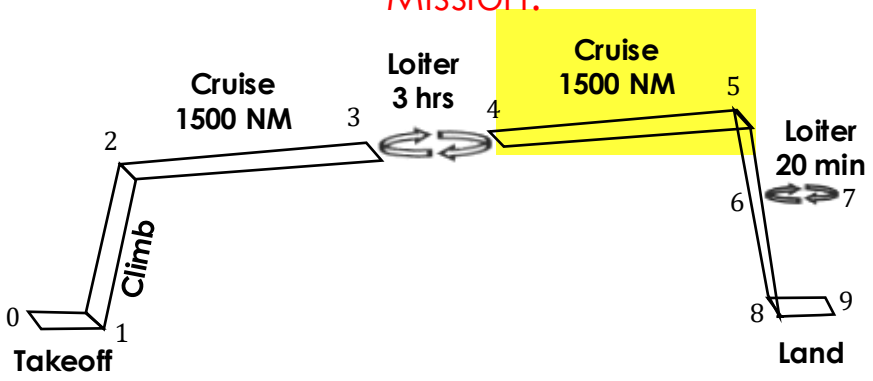
$C = 0.4\frac{1}{hr} = 0.000111\frac{1}{s}$

$\left(\frac{L}{D}\right) = \left(\frac{L}{D}\right)_{max} = 16$

	Cruise	Loiter
Jet	$0.866\,(L/D)_{max}$	$(L/D)_{max}$

Design Example: ASW Aircraft

Mission:



- The aircraft must cruise at 0.6 Mach Number.
- Equivalent combined aspect ratio of 7.

$$\frac{W_5}{W_4} = 0.853, same\ as, \frac{W_3}{W_2}$$

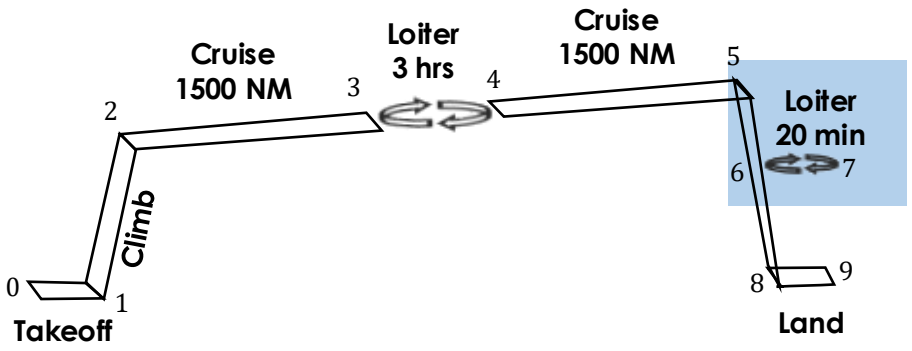
$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

$$1 - \frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}}$$ Mission fuel fraction

$$\frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}} = \frac{W_9}{W_0} = (0.97)(0.985)\ (0.853)(0.9277)\frac{W_5}{W_4}\frac{W_7}{W_6}(0.995)$$

Design Example: ASW Aircraft

Mission:



- The aircraft must cruise at 0.6 Mach Number.
- Equivalent combined aspect ratio of 7.

$$\frac{W_7}{W_6} = e^{\frac{-EC}{(L/D)}}$$

$E = 20\ mins = 1200\ s$

$C = 0.4\frac{1}{hr} = 0.000111\frac{1}{s}$

$\left(\frac{L}{D}\right) = \left(\frac{L}{D}\right)_{max} = 16$

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

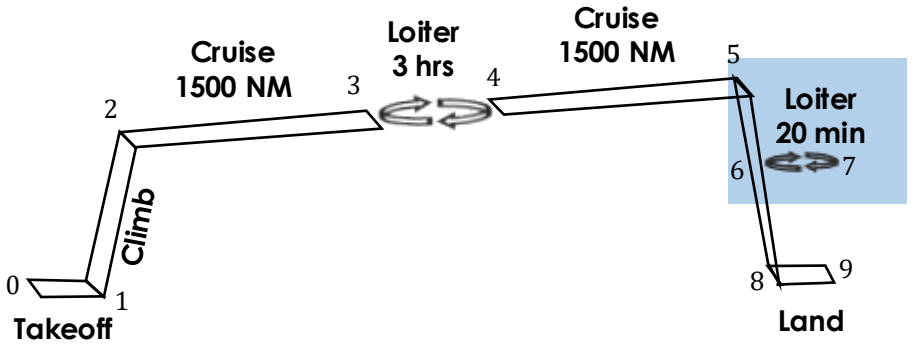
$1 - \frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}}$ Mission fuel fraction

$$\frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}} = \frac{W_9}{W_0}$$

$$= (0.97)(0.985)\ (0.853)(0.9277)(0.853)\ \frac{W_7}{W_6}(0.995)$$

Design Example: ASW Aircraft

Mission:



- The aircraft must cruise at 0.6 Mach Number.
- Equivalent combined aspect ratio of 7.

$$\frac{W_7}{W_6} = e^{\frac{-EC}{(L/D)}} = e^{\frac{-1200(0.00011)}{(16)}} = 0.9917$$

$$E = 20\ mins = 1200\ s$$

$$C = 0.4\frac{1}{hr} = 0.000111\frac{1}{s}$$

$$\left(\frac{L}{D}\right) = \left(\frac{L}{D}\right)_{max} = 16$$

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

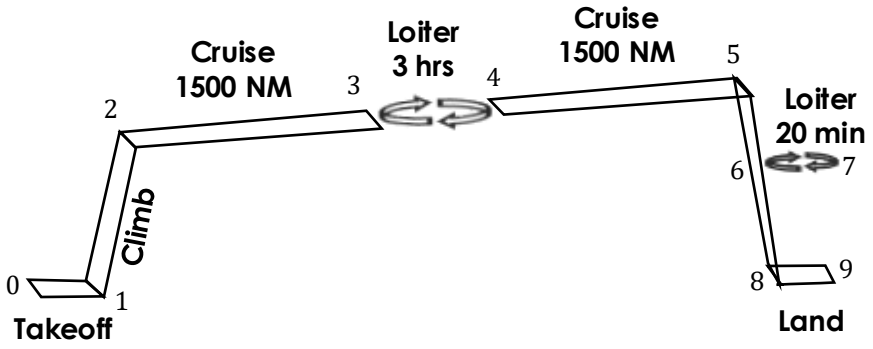
$$1 - \frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}}$$ Mission fuel fraction

$$\frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}} = \frac{W_9}{W_0}$$

$$= (0.97)(0.985)\ (0.853)(0.9277)(0.853)\ \frac{W_7}{W_6}(0.995)$$

Design Example: ASW Aircraft

Mission:



- The aircraft must cruise at 0.6 Mach Number.
- Equivalent combined aspect ratio of 7.

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

$$1 - \frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}}$$ Mission fuel fraction

$$\frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}} = \frac{W_9}{W_0}$$

$$= (0.97)(0.985)(0.853)(0.9277)(0.853)(0.9917)(0.995)$$

$$= 0.635$$

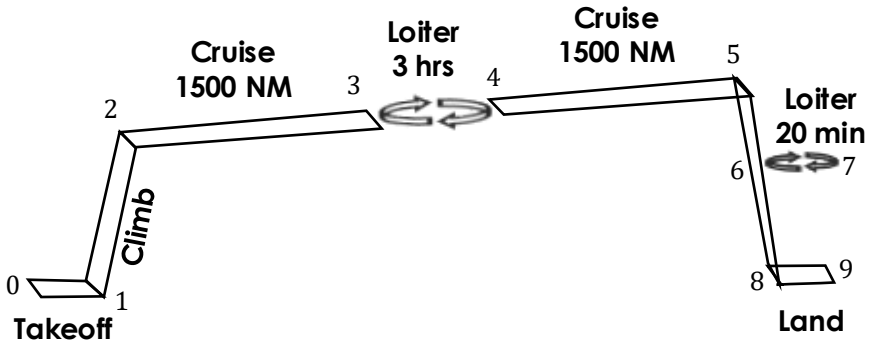
$$1 - \frac{\boldsymbol{W_{final}}}{\boldsymbol{W_{initial}}} = 1 - 0.635 = 0.365$$

If %6 reserve fuel:

$$\frac{W_f}{W_0} = 1.06\left(1 - \frac{W_x}{W_0}\right) = 0.387$$

Design Example: ASW Aircraft

Mission:



- The aircraft must cruise at 0.6 Mach Number.
- Equivalent combined aspect ratio of 7.

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

$$W_0 = \frac{10800}{1 - (0.387) - (0.93 W_0^{-0.07})}$$

W_0 *Guess*	$\frac{W_e}{W_0}$	W_0 *Calculated*
50,000	0.4361	61,057
60,000	0.4305	59,191
59,200	0.4309	59,328
59,300	0.4309	59,311
59,310	0.4309	59,309.6

Design Example: ASW Aircraft – Trade Studies

An important part of conceptual design is the evaluation and refinement, with the customer, of the design requirements:

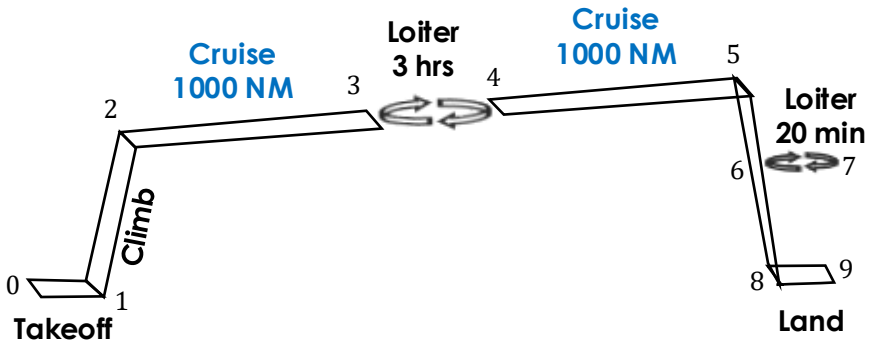
The required range: 1500 nm is probably less than the customer would really like.

A "range trade" can be calculated to determine the increase in design takeoff gross weight if the required range is increased.

This is done by recalculating the weight fractions for the cruise mission segments, using arbitrary selected ranges: 1000 nm, and 2000 nm.

Design Example: ASW Aircraft – Trade Studies

1000 nm Range:



$$\frac{W_3}{W_2} = e^{\frac{-RC}{V(L/D)}}$$

$$V = 569.9\ ft/s$$

$$C = 0.5\frac{1}{hr} = 0.0001389\frac{1}{s}$$

$$R = 1000nm = 6076000\ ft$$

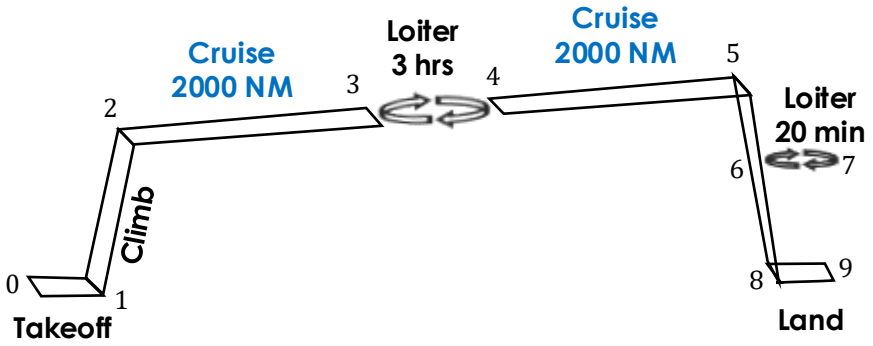
$$\frac{W_3}{W_2} = 0.899$$

$$\frac{W_9}{W_0} = 0.7069$$

$$\frac{W_f}{W_0} = 1.06(1 - 0.7069) = 0.3107$$

Iteration gives: W_0 = 43,498 lb

2000 nm Range:



$$\frac{W_3}{W_2} = e^{\frac{-RC}{V(L/D)}}$$

$$V = 569.9\ ft/s$$

$$C = 0.5\frac{1}{hr} = 0.0001389\frac{1}{s}$$

$$R = 2000nm = 12152000\ ft$$

$$\frac{W_3}{W_2} = 0.8082$$

$$\frac{W_9}{W_0} = 0.5713$$

$$\frac{W_f}{W_0} = 0.4544$$

Iteration gives: W_0 = 85,911 lb

Design Example: ASW Aircraft – Trade Studies

Payload Trade

$$W_{payload} = 5000\ lb$$

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

$$W_0 = \frac{5800}{0.613\ - 0.93 W_0^{-0.07}}$$

Iteration gives: $W_0 = 34{,}966$ lb

$$W_{payload} = 15{,}000\ lb$$

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

$$W_0 = \frac{15800}{0.613\ - 0.93 W_0^{-0.07}}$$

Iteration gives: $W_0 = 82{,}350$ lb

$$W_{payload} = 20{,}000\ lb$$

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \left(\frac{W_f}{W_0}\right) - \left(\frac{W_e}{W_0}\right)}$$

$$W_0 = \frac{20800}{0.613\ - 0.93 W_0^{-0.07}}$$

Iteration gives: $W_0 = 104{,}600$ lb

Design Example: ASW Aircraft

Note on Empty Weight Fraction:

- The takeoff gross weight calculations above have thus implicitly assumed that the new aircraft would also be built of aluminum.
- To determine the effect of building the aircraft out of composite materials, the designer must adjust the empty weight equation.
- This can be approximated by taking %95 of the empty-weight fraction obtained for a metal aircraft.

$$\frac{W_e}{W_0} = 0.95\ (A\ W_0^c K_{vs})$$